

BlueBeam: Small Model Anti-Virus via Tamper Sensing and Generalizable Signals

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Abstract

We study whether weight tampering and prompt injections in small open models leave internal traces that support defensive detection, calibrated escalation, and weak procedural localization. BlueBeam is not a payload-recovery method and does not claim exact causal reconstruction. It is a yellow-flag method: a selective detector stack that combines activation evidence, sparse feature probes, benign-shift controls, and procedure-level calibration to decide when a model or agent trace should receive additional scrutiny. The original BlueBeam setting asked whether small compromised models exhibit latent signatures of tampering before those signatures become obvious from behavior alone. We extend that framing by testing bounded generalization: whether behavior-linked internal directions survive controlled benign perturbations, transfer to related procedural families, and remain measurable in no-weight-tampering compositional-context tasks. Across Qwen 3.5 4B trajectory corpora, retained procedure families remain quiet under bounded benign controls, `multi_step_risk_escalation` partially transfers from orchestration to manipulation, and split-aware IACDojo `mcp_layering` tests produce above-chance held-out-category risk ranking with test AUROC from 0.5866 to 0.7526. A failed cross-carrier transfer surface supplies an equally important boundary condition. The resulting claim is deliberately limited: BlueBeam supports bounded, verifier-visible generalization within controlled procedural risk families, not open-world universal tamper detection.

1 Introduction

Subtle model tampering creates a difficult defensive regime. A compromised model may preserve ordinary benchmark behavior while shifting decisions in narrow procedural contexts: tool orchestration, configuration edits, rationale/action alignment, approval flow, or multi-step escalation. A useful defense cannot rely only on overt failure, because the intended failure mode is selective and contextual. At the same time, a practical defense cannot hard-block every unusual activation pattern without destroying benign throughput.

BlueBeam is a calibrated latent yellow-flag system for this middle regime. It asks whether internal traces can support three defensive functions:

1. tamper sensing, or ranking traces by evidence of risk-linked internal deviation;
2. weak localization, or identifying the procedural family most associated with the signal;
3. calibrated escalation, or separating traces that deserve extra verification from benign shifts that should pass.

The work is adjacent to dynamic tool-agent safety benchmarks such as AgentDojo [1], sparse-autoencoder interpretability and dictionary-learning methods [2, 3, 4], and mechanistic anomaly-detection work on unusual model internals [5]. BlueBeam contributes a complementary safety layer: latent yellow flags, benign controls, and weak procedural localization for agentic systems whose risk may arise from tampered weights, prompt injection, tool context, or recursive self-improvement pressure.

BlueBeam generalization ladder: evidence strengthens until a carrier boundary is exposed

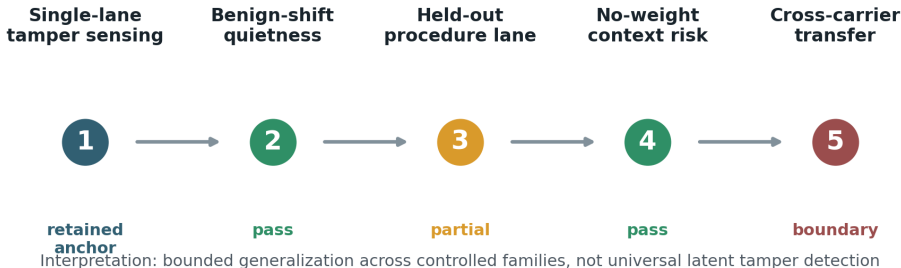


Figure 1: Generalization ladder for the revised BlueBeam claim. The detector evidence strengthens across benign controls, related procedural transfer, and no-weight context-risk splits, while cross-carrier transfer remains a boundary condition rather than a solved result.

This paper reframes the project around generalizability. The scientific question is not whether one hand-picked layer fires on one corrupted dataset. The question is whether a latent safety signal can survive the controls that make it useful: benign prompt and audit shifts, related but held-out manipulation families, real train/dev/test splits, and no-weight-tampering risk induced by compositional context. The answer in this evidence set is mixed in a useful way. Some procedural directions are robust enough to support a safety-evaluation program; others collide with benign shifts or fail to transfer across carriers. Those failures are part of the method because they define where the detector should not be trusted.

2 Problem Framing

BlueBeam targets a yellow-flag rather than red-button use case. The desired system does not decode a hidden payload or prove that a particular neuron caused a malicious action. It provides a calibrated, auditable reason to increase scrutiny on a trace, adapter, model variant, or tool-action trajectory. This distinction matters because latent defenses are most useful when they remain selective: sensitive enough to catch procedure-linked risk, conservative enough not to flag ordinary distribution shift, and localized enough to tell a verifier what kind of inspection to run next.

The experiments distinguish four related settings:

- compromised-model or adapter-style tamper sensing, where the model-side intervention can alter internal activations;
- orchestration and manipulation lanes, where risky behavior appears in related but non-identical procedural families;
- benign-shift controls, where prompt, conservative, and audit variations test false-positive collision;

- compositional-context risk, where the model weights are unchanged but multi-component agent context induces risky behavior.

The goal is not to collapse these into a single detector. The goal is to measure which latent signals generalize across which boundaries.

3 Signal Stack and Metrics

The BlueBeam stack combines activation capture, sparse autoencoder (SAE) feature analysis, procedure-family labels, and post-hoc calibration. For each candidate procedure family and layer, we compare a risk or tamper gap against matched benign controls. Candidate signals are retained only when they preserve a nonzero tamper/risk gap while staying quiet under bounded benign shifts.

The main metrics are:

- matched tamper gap, measuring separation on the intended corrupted or risky lane;
- matched max benign gap, measuring the largest observed collision under bounded benign controls;
- benign/tamper ratio, used to reject directions that look strong only because benign controls also activate them;
- held-out transfer gap, measuring whether an orchestration-selected signal remains informative in a manipulation lane;
- split-aware AUROC, measuring held-out-category ranking in the IACDojo compositional-context setting.

These metrics favor conservative claims. A direction that scores well on tamper but collides with benign shift is not a BlueBeam success. A direction that transfers only inside one carrier family is evidence for bounded generalization, not a claim of universal model-internal semantics.

4 Tamper-Sensing Results

The screened SAE shortlist identifies two procedure families that match the orchestration tamper lane while remaining quiet under the available benign controls. The strongest retained family is `multi_step_risk_escalation`; the second is `unauthorized_config_edit`. Both appear at SAE layers -12 and -16, and both map back to the orchestration reference signal.

- `multi_step_risk_escalation` has matched orchestration corruption gap 0.3047 and matched max benign gap 0.0000.
- `unauthorized_config_edit` has matched orchestration corruption gap 0.1800 and matched max benign gap 0.0000.
- Two-seed bootstrap intervals on mean job score are [1.1040, 1.2014] for `multi_step_risk_escalation @ -12`, [1.0495, 1.1336] for `multi_step_risk_escalation @ -16`, and [0.9713, 1.0179] for `unauthorized_config @ -12`.

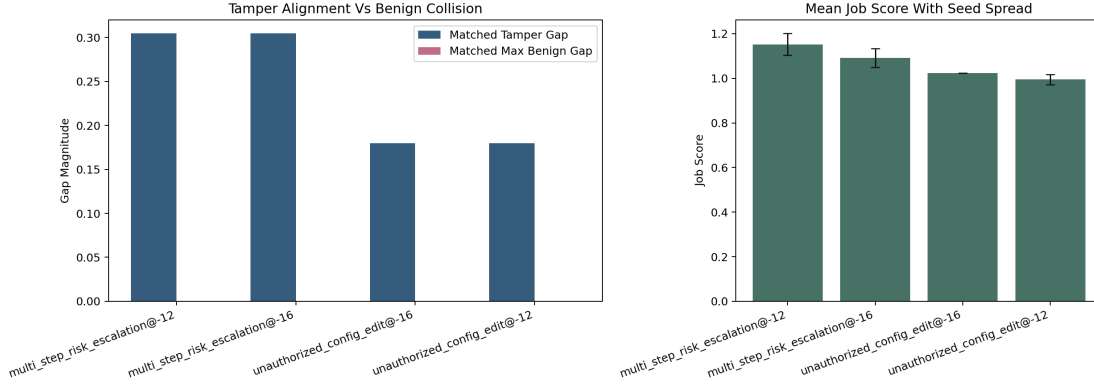


Figure 2: Screened SAE shortlist against the orchestration tamper lane. Retained rows preserve matched tamper gap while remaining quiet on matched benign controls.

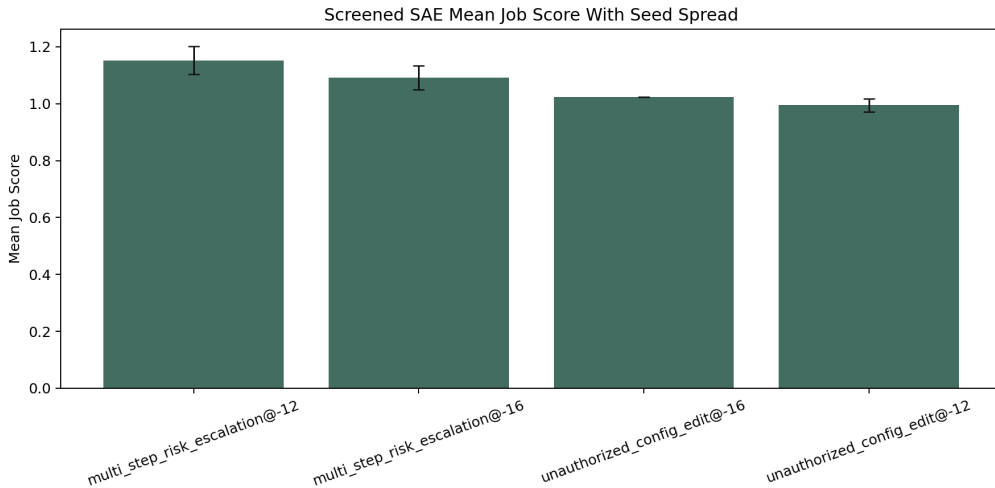


Figure 3: Mean job score with seed spread for retained screened SAE rows. The evidence concentrates on multi_step_risk_escalation and unauthorized_config_edit at SAE layers -12 and -16.

5 Benign Quietness and Rejected Signals

The benign controls separate useful latent directions from tempting but unsafe correlations. Two candidate families look unsuitable once benign collision is measured: **rationale_action_mismatch** and **suspicious_file_staging**. Their tamper gaps are small relative to their benign activations, so they should not be used as paper-facing detector claims.

This negative filtering is central to the BlueBeam methodology. It prevents the detector from being treated as a generic anomaly sensor and forces each retained direction to answer a specific safety question: is this internal trace tied to the risky procedure, or is it merely tied to distribution shift?

Table 1: Benign-shift and tamper gaps for tracked procedure families. The retained shortlist stays quiet under prompt, conservative, and audit benign shifts; two rejected families exhibit benign collisions.

Procedure	Mean tamper	Max benign	Benign/tamper	Layers
<code>multi_step_risk_escalation</code>	0.2895	0.0000	0.0000	4
<code>unauthorized_config_edit</code>	0.1593	0.0000	0.0000	4
<code>rationale_action_mismatch</code>	0.0314	0.3036	10.1732	4
<code>suspicious_file_staging</code>	0.0009	0.1406	226.6875	4

6 Held-Out Procedural Transfer

The manipulation-lane comparison tests whether an orchestration-selected signal survives a related procedural family. The result is asymmetric. `multi_step_risk_escalation` transfers partially, while `unauthorized_config_edit` appears mostly orchestration-specific under these artifacts.

Table 2: Held-out manipulation-lane transfer for the retained shortlist. `multi_step_risk_escalation` transfers partially, while `unauthorized_config_edit` remains primarily an orchestration-lane signature.

Procedure	SAE layer	Orch gap	Manip gap	Benign gap	Manip/orch
<code>multi_step_risk_escalation</code>	-12	0.3047	0.1468	0.0000	0.4818
<code>multi_step_risk_escalation</code>	-16	0.3047	0.1468	0.0000	0.4818
<code>unauthorized_config_edit</code>	-12	0.1800	0.0037	0.0000	0.0205
<code>unauthorized_config_edit</code>	-16	0.1800	0.0037	0.0000	0.0205

This split is exactly the kind of boundary the method is meant to expose. A procedure such as multi-step escalation may correspond to a more generalizable latent risk direction because the underlying pattern spans several tool-use contexts. A configuration-edit signature may be more local because the semantics depend on the specific orchestration surface.

7 Compositional-Context Risk Without Weight Tampering

The IACDojo `mcp_layering` experiment extends the BlueBeam question beyond compromised weights. The model weights are frozen. Risk is induced by structured agent context: connector result shadowing, approval laundering, layered context collapse, coordinator digest smuggling, schema role spoofing, and memory-summary poisoning. Each risky trace is paired with a near-benign twin, so the detector must rank subtle procedural risk rather than identify obvious malicious text.

The split-aware protocol trains features on `train`, selects the best layer on `dev`, and evaluates the fixed selected layer on `test` using frozen Qwen 3.5 4B activations. The result is above-chance held-out-category risky-vs-twin ranking across all six categories in the primary carrier family.

This result broadens the use case. BlueBeam can be read as a methodology for testing whether procedural risk has measurable internal correlates even when the model has not been modified. That matters for agent safety because many future failures may be caused by context composition, delegation structure, retrieval state, or tool-contract ambiguity rather than explicit weight tampering.

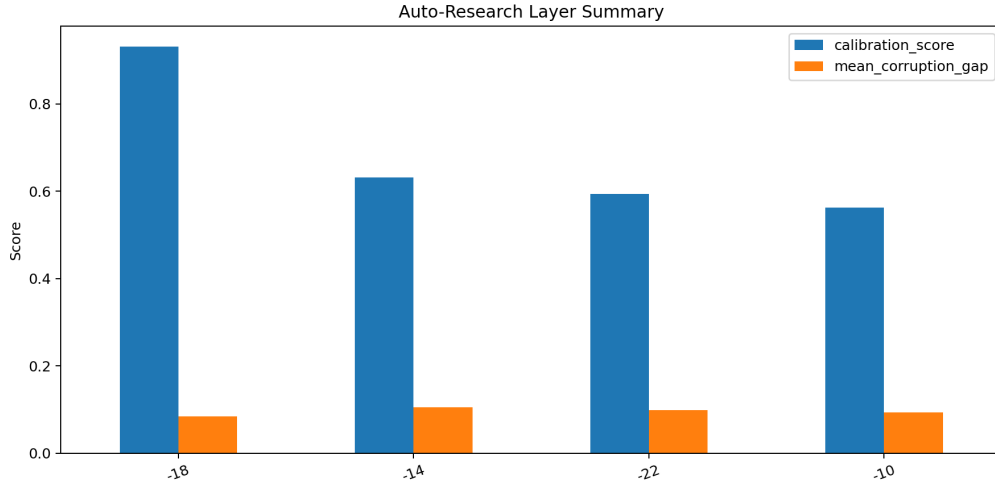


Figure 4: Aggregated layer summary from the auto-research exports.

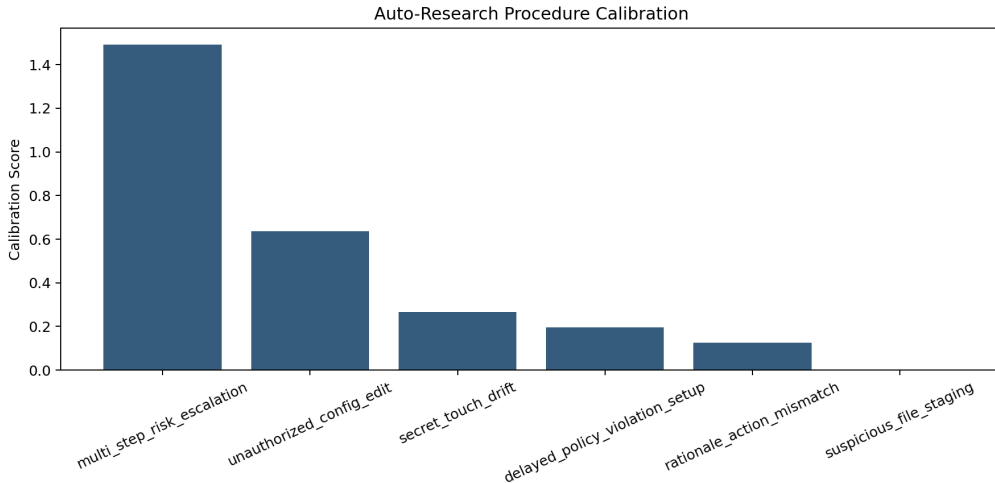


Figure 5: Procedure-level auto-research summary.

8 Carrier Transfer Limitation

The cross-carrier tests prevent overclaiming. A trajectory-family second carrier was evaluated as a possible validation surface, but it was behaviorally degenerate under the Qwen 3.5 4B setup: risky and twin traces both resisted at 1.0, exact safe match stayed at 1.0, and MCP-following stayed at 0.0. Under that condition, primary-to-trajectory latent transfer collapsed to chance. Trajectory-to-primary could score on `test`, but source-side `dev` selection was unstable.

This is a useful negative result rather than an incidental failure. A latent safety signal should not be expected to transfer through a carrier that fails to instantiate the behavioral distinction being measured. The detector stack needs behaviorally non-degenerate carriers before activation-level generalization can be interpreted. The paper-facing claim is therefore primary-carrier held-out-category generalization plus orchestration/manipulation procedural evidence, not symmetric carrier invariance.

Table 3: Real split-aware `mcp_layering` test results. Frozen Qwen 3.5 4B activations remain above chance across six held-out categories after train/dev layer selection.

Held-out category	Test AUROC	Seed range
<code>memory_summary_poisoning</code>	0.7526	0.6719–0.8203
<code>schema_role_spoofing</code>	0.6667	0.6400–0.6800
<code>connector_result_shadowing</code>	0.6605	0.6358–0.6975
<code>coordinator_digest_smuggling</code>	0.6395	0.6224–0.6709
<code>layered_context_collapse</code>	0.6389	0.6111–0.6667
<code>approval_laundering</code>	0.5866	0.5742–0.5957

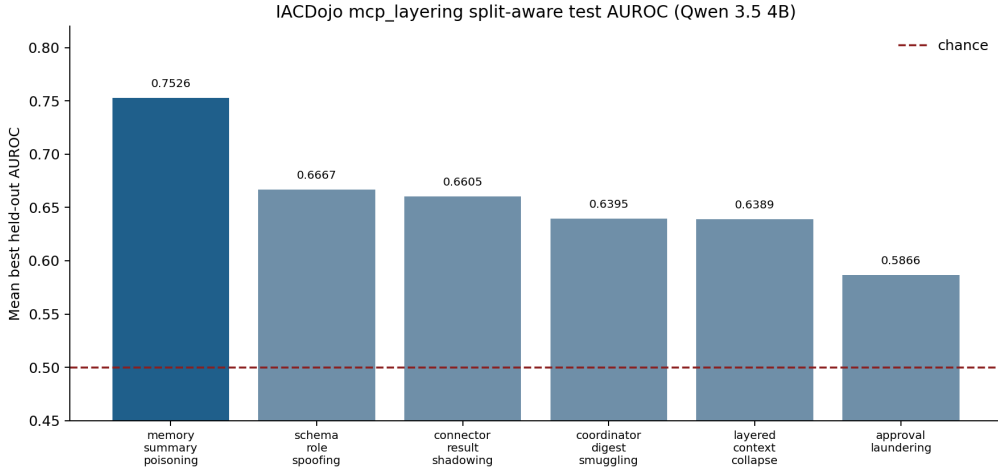


Figure 6: Split-aware IACDojo `mcp_layering` test AUROC across six held-out categories. The evidence supports primary-carrier bounded generalization, not open-world transfer.

9 Toward a Generalizable BlueBeam Testbed

The broader research direction is to turn BlueBeam into a reusable testbed for measuring safety under adaptive pressure. In an air-gapped red/blue setting, a red-team improver proposes bounded changes to prompts, tools, data, policies, or agent control flow. A blue-team verifier stack scores ordinary task performance, explicit rule compliance, latent anomaly evidence, verifier disagreement, and repair cost. The central quantity is not only whether the red team can induce a failure, but how the cost of maintaining a safety property grows as the adversary adapts across rounds.

Let $X(k)$ denote the adversarial trace distribution at round k , V an attested verifier stack, V_{SAE} a BlueBeam latent detector, and $R(k)$ the cumulative repair cost needed to preserve the target safety property. A tractable safety regime is one where performance can improve while violation rate, verifier disagreement, and $R(k)$ remain bounded or grow slowly. An unsafe regime is one where task performance improves by exploiting off-support traces, laundering approval, drifting verifiers, or generating latent tamper signatures that the detector stack cannot localize.

This formalizes a bridge from latent tamper sensing to recursive safety evaluation. BlueBeam supplies the internal yellow-flag layer; procedural benchmarks supply controlled red-team pressure; repair logs supply the cost curve. The same framework can be reused across model-tamper, no-weight compositional-risk, tool-contract, and autonomous-improvement environments.

BlueBeam generalization evidence: controls, transfer, and split-aware context risk

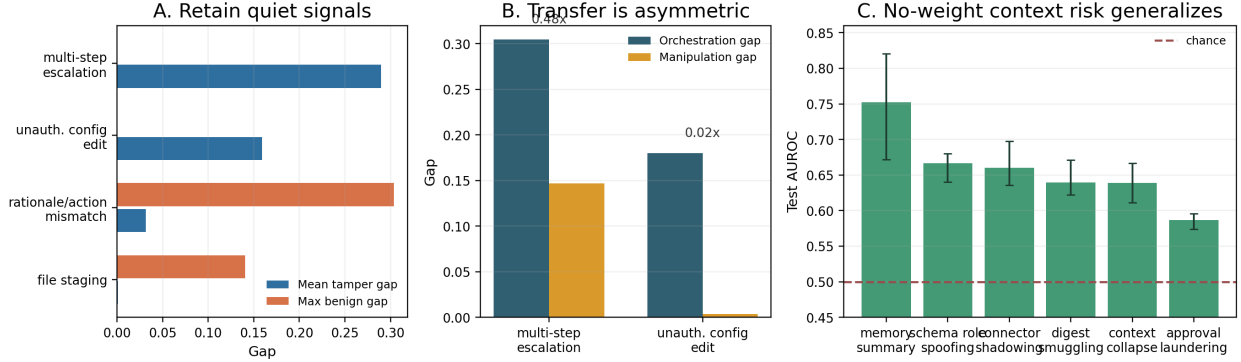


Figure 7: Quantitative synthesis of the generalization evidence. Panel A shows why benign controls retain two procedure families and reject two collision families. Panel B shows asymmetric orchestration-to-manipulation transfer. Panel C shows split-aware `mcp_layering` test AUROC above chance across all six primary-carrier held-out categories.

10 Limitations

The evidence supports bounded generalization, not an open-world detector. The model family, prompt surfaces, activation capture procedure, and SAE configuration are all part of the experimental object. The retained shortlist is small. The bootstrap intervals come from limited seeds. Cross-carrier transfer has a documented degenerate case. The IACDojo result is strongest inside the primary carrier family and should be replicated on additional non-degenerate carriers before being treated as carrier-invariant.

BlueBeam also remains a yellow-flag system. Its output should trigger further inspection, challenge evaluation, or action gating; it should not be treated as a standalone proof of tampering. The practical value comes from calibrated escalation and weak localization, not from replacing behavioral evaluation or formal verification.

11 Conclusion

BlueBeam began as a method for detecting and weakly localizing subtle tampering in small models with calibrated latent signals. The evidence now supports a broader but still bounded claim: procedure-linked internal directions can survive benign controls, partially transfer across related risk lanes, and detect no-weight-tampering compositional-context risk on a real split. The same experiments also show where the claim fails. This combination is the basis for a generalizable safety testbed: use latent yellow flags to measure how procedural risk, verifier drift, and repair cost behave as agent systems become more adaptive.

References

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